

MAI-Voice-2.1-Flash Model Card

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Overview

MAI-Voice-2.1-Flash is a prompted text-to-speech (TTS) model built in-house by the Microsoft AI team for fast, low-latency generation. It delivers high-fidelity, natural, and expressive speech across 23 languages while being optimized for real-time responsiveness, preserving human-like intonation, rhythm, and emotional nuance. Developers can control tone, emotion, and delivery through SSML, making it ideal for voice agents, call center agents, assistants, and other interactive scenarios where speed is critical.

Voice can be configured using:

- Curated voice library (licensed voices designed to work straight out of the box)
- Voice Cloning via short audio clips (5-60 seconds), and the model matches the voice instantly

Key capabilities

- Natural and expressive voice synthesis with realistic intonation, rhythm, and emotional range.
- Low latency optimized for real-time use cases like voice assistants and call center agents.
- High-fidelity, high-clarity voice output.
- Multilingual support across 23 languages.
- One voice across all supported languages: a single voice identity is rendered consistently in every supported language.
- Fine-grained control of tone and delivery: shape emotion, tone, and delivery at the turn/sentence level, including different speaking styles via SSML.
- Streaming output, enabling barge-in and interruption handling for live dialogue.
- State-of-the-art instant Voice Cloning: provide a short audio clip (5-60 seconds) and the model clones the voice instantly, with no fine-tuning required. Prompt quality significantly impacts output, with best results from natural, conversational delivery and moderate energy levels. Access requires Microsoft approval and guardrails are in place to avoid misuse.
- Multi-speaker support for dialogue and conversational scenarios.



Voice Cloning safeguards

To ensure Voice Cloning is compliant, we adhere to Microsoft's Responsible AI policies and apply the following steps to ensure voice cloning is conducted lawfully.

Steps to Access Personal Voice (Voice Cloning) with MAI-Voice-2.1-Flash:

- Apply for gated access: Submit a request via the Azure AI Custom Neural Voice and Custom Avatar Limited Access Review.
- Access personal voice APIs: Once approved, use the APIs available at `cognitive-services-speech-sdk/samples/custom-voice`
- Upload audio consent and prompt: Provide a recorded audio consent statement from the voice talent along with the required prompt to create a personal voice profile. For more information, read about [Azure Speech consent statement](#).
- Synthesize text using the created voice

Use cases

Key use cases

- Call Center Agents: Support real-time agent workflows with natural, expressive voice output.
- Virtual Assistants and Chatbots: Power conversational agents across apps and devices with natural voices.
- IVR Systems: Enable natural, expressive call center interactions.
- Live Customer Support: Deliver responsive spoken replies during live interactions.
- Accessibility Features: Provide narration for visually impaired users and assistive voice technologies.
- Educational Experiences: Build interactive learning content with expressive narration.
- Public Announcements: Deliver clear, engaging voice output for public information systems.

Pricing

\$15 per 1M characters

Technical specs

Input formats

Transcript text with an optional speaker prompt for Voice Cloning (with 5-60s of audio), plus optional emotion tags like 'joyful' or 'relieved' via SSML.



SSML examples

Basic SSML Example (Harper):

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Harper:MAI-Voice-2.1-Flash'>
    Hello world, this is awesome!
  </voice>
</speak>
```

Basic SSML Example (Grant):

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Grant:MAI-Voice-2.1-Flash'>
    Hello world, this is awesome!
  </voice>
</speak>
```

Multilingual SSML Example (Harper):

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Harper:MAI-Voice-2.1-Flash'> How are you? </voice>
  <voice name='hu-HU-Harper:MAI-Voice-2.1-Flash'> Hogy vagy? </voice>
  <voice name='vi-VN-Harper:MAI-Voice-2.1-Flash'> Bạn khỏe không? </voice>
  <voice name='es-MX-Harper:MAI-Voice-2.1-Flash'> ¿Cómo estás? </voice>
</speak>
```

Multilingual SSML Example (Grant):

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Grant:MAI-Voice-2.1-Flash'>I am fine, thank you</voice>
  <voice name='hu-HU-Grant:MAI-Voice-2.1-Flash'>Jól vagyok, köszönöm.</voice>
  <voice name='vi-VN-Grant:MAI-Voice-2.1-Flash'>Tôi ổn, cảm ơn bạn.</voice>
  <voice name='es-MX-Grant:MAI-Voice-2.1-Flash'>Estoy bien gracias.</voice>
</speak>
```

Expressive Control with SSML `mstts:express-as`:

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:mstts='http://www.w3.org/2001/mstts'
xml:lang='en-US'>
  <voice name='en-US-Grant:MAI-Voice-2.1-Flash'>
    <mstts:express-as style="excited">
      Hello world, this is awesome!
    </mstts:express-as>
  </voice>
</speak>
```

Output

Users can specify the desired output format. More information [here](#).



Supported languages

23 languages:

English (US), English (Australia), Italian, French, German, Hindi, Spanish (Spain), Spanish (Mexico), Portuguese (Brazil), Portuguese (Portugal), Korean, Chinese (Simplified), Turkish, Russian, Thai, Dutch, Romanian, Hungarian, Czech, Danish, Finnish, Indonesian, Polish, Swedish, Norwegian (Bokmål), Vietnamese.

Distribution

Ready to integrate? Explore the full [MAI-Voice documentation](#) to get started.

MAI-Voice-2.1 is available via the Azure Speech SDK and REST APIs:

- Speech SDK: Integrate TTS capabilities directly into applications using Azure's Speech SDK, available for platforms including .NET, Python, Java, JavaScript, and C++.
- REST API: Access TTS functionality via a public, subscription-based API for flexible integration into web services, mobile apps, and backend systems.

MAI-Voice-2.1-Flash can be accessed globally. The model is currently served from the following Azure regions to which the requests are routed:

- East Asia
- France Central
- South East Asia
- East US
- Canada Central
- East US 2
- West US
- West Europe
- North Europe
- West US 2
- West US 3
- Central India
- Sweden Central
- Japan East